

Table 1: Quantitative results (mean $\pm$ std) on time-series datasets across sequence lengths. The best results are indicated as **bold** for each metric and dataset.

Length	Dataset	Method	Disc. ($\downarrow$)	Pred. ($\downarrow$)	cFID ($\downarrow$)	Corr. ($\downarrow$)
128	Energy	TimeVAE	0.492 $\pm$ 0.022	79.276 $\pm$ 6.222	2.009	0.018 $\pm$ 0.226
		TimeGAN	0.500 $\pm$ 0.000	74.723 $\pm$ 8.445	4.099	0.014 $\pm$ 0.076
		Diffusion-TS	0.500 $\pm$ 0.000	79.018 $\pm$ 9.238	2.396	0.002 $\pm$ 0.212
		TimeDiff	0.484 $\pm$ 0.025	77.426 $\pm$ 22.354	2.188	0.017 $\pm$ 0.181
		ImagenI2R	0.488 $\pm$ 0.017	83.615 $\pm$ 18.807	2.014	0.002 $\pm$ 0.091
		ImagenTime	0.488 $\pm$ 0.021	67.624 $\pm$ 21.872	2.164	0.003 $\pm$ 0.116
		TSFlow	0.500 $\pm$ 0.000	76.693 $\pm$ 7.682	2.208	0.003 $\pm$ 0.018
		FLOW-ITS	0.463 $\pm$ 0.018	68.870 $\pm$ 8.902	1.848	0.002 $\pm$ 0.026
	ETTh	TimeVAE	0.500 $\pm$ 0.000	1.250 $\pm$ 0.168	1.210	0.030 $\pm$ 0.240
		TimeGAN	0.497 $\pm$ 0.007	1.416 $\pm$ 0.114	1.390	0.005 $\pm$ 0.072
		Diffusion-TS	0.500 $\pm$ 0.000	1.214 $\pm$ 0.126	0.862	0.026 $\pm$ 0.212
		TimeDiff	0.474 $\pm$ 0.046	0.979 $\pm$ 0.463	0.961	0.031 $\pm$ 0.141
		ImagenI2R	0.481 $\pm$ 0.031	1.145 $\pm$ 0.526	1.488	0.059 $\pm$ 0.166
		ImagenTime	0.405 $\pm$ 0.077	0.466 $\pm$ 0.256	0.904	0.003 $\pm$ 0.167
		TSFlow	0.495 $\pm$ 0.010	1.470 $\pm$ 0.193	0.915	0.017 $\pm$ 0.133
		FLOW-ITS	0.328 $\pm$ 0.085	0.524 $\pm$ 0.203	0.817	0.001 $\pm$ 0.081
	LMT	TimeVAE	0.500 $\pm$ 0.000	32.250 $\pm$ 18.253	3.484	0.070 $\pm$ 0.309
		TimeGAN	0.465 $\pm$ 0.037	0.850 $\pm$ 0.379	0.554	0.057 $\pm$ 0.138
		Diffusion-TS	0.500 $\pm$ 0.000	42.965 $\pm$ 14.793	4.459	0.065 $\pm$ 0.346
		TimeDiff	0.500 $\pm$ 0.000	30.006 $\pm$ 14.972	5.150	0.009 $\pm$ 0.321
		ImagenI2R	0.425 $\pm$ 0.079	2.880 $\pm$ 0.929	0.828	0.093 $\pm$ 0.292
		ImagenTime	0.500 $\pm$ 0.000	11.722 $\pm$ 3.420	2.653	0.046 $\pm$ 0.333
		TSFlow	0.437 $\pm$ 0.094	0.516 $\pm$ 0.066	0.554	0.112 $\pm$ 0.095
		FLOW-ITS	0.267 $\pm$ 0.097	0.037 $\pm$ 0.013	0.125	0.050 $\pm$ 0.386
256	Energy	TimeVAE	0.496 $\pm$ 0.009	79.488 $\pm$ 7.629	1.747	0.017 $\pm$ 0.230
		TimeGAN	0.500 $\pm$ 0.000	71.728 $\pm$ 10.134	1.947	0.011 $\pm$ 0.083
		Diffusion-TS	0.496 $\pm$ 0.009	80.322 $\pm$ 8.800	1.730	0.011 $\pm$ 0.190
		TimeDiff	0.490 $\pm$ 0.016	61.650 $\pm$ 23.826	3.491	0.022 $\pm$ 0.153
		ImagenI2R	0.487 $\pm$ 0.022	63.600 $\pm$ 17.849	1.854	0.001 $\pm$ 0.014
		ImagenTime	0.487 $\pm$ 0.020	60.576 $\pm$ 19.843	1.905	0.046 $\pm$ 0.140
		TSFlow	0.496 $\pm$ 0.009	65.836 $\pm$ 5.504	1.707	0.005 $\pm$ 0.012
		FLOW-ITS	0.468 $\pm$ 0.025	60.890 $\pm$ 6.645	1.573	0.005 $\pm$ 0.046
	ETTh	TimeVAE	0.500 $\pm$ 0.000	1.405 $\pm$ 0.130	1.081	0.013 $\pm$ 0.214
		TimeGAN	0.496 $\pm$ 0.010	1.274 $\pm$ 0.117	1.356	0.010 $\pm$ 0.051
		Diffusion-TS	0.496 $\pm$ 0.010	1.318 $\pm$ 0.334	0.802	0.009 $\pm$ 0.186
		TimeDiff	0.463 $\pm$ 0.056	1.157 $\pm$ 0.575	0.669	0.032 $\pm$ 0.123
		ImagenI2R	0.470 $\pm$ 0.038	1.092 $\pm$ 0.555	1.063	0.011 $\pm$ 0.187
		ImagenTime	0.404 $\pm$ 0.068	0.550 $\pm$ 0.298	0.847	0.030 $\pm$ 0.163
		TSFlow	0.471 $\pm$ 0.054	2.004 $\pm$ 0.510	1.507	0.015 $\pm$ 0.116
		FLOW-ITS	0.368 $\pm$ 0.093	0.532 $\pm$ 0.256	0.746	0.006 $\pm$ 0.096
	LMT	TimeVAE	0.500 $\pm$ 0.000	25.673 $\pm$ 14.495	2.807	0.035 $\pm$ 0.281
		TimeGAN	0.492 $\pm$ 0.011	1.355 $\pm$ 0.471	3.250	0.059 $\pm$ 0.062
		Diffusion-TS	0.500 $\pm$ 0.000	19.811 $\pm$ 11.199	1.910	0.040 $\pm$ 0.270
		TimeDiff	0.500 $\pm$ 0.000	20.179 $\pm$ 10.045	3.874	0.012 $\pm$ 0.188
		ImagenI2R	0.475 $\pm$ 0.038	0.390 $\pm$ 0.409	0.811	0.009 $\pm$ 0.270
		ImagenTime	0.500 $\pm$ 0.000	19.370 $\pm$ 10.346	2.644	0.079 $\pm$ 0.204
		TSFlow	0.405 $\pm$ 0.104	0.566 $\pm$ 0.021	1.666	0.061 $\pm$ 0.062
		FLOW-ITS	0.375 $\pm$ 0.067	0.173 $\pm$ 0.176	0.607	0.045 $\pm$ 0.292

Table 2: Ablation study on Energy, ETTh and LMT at a sequence length of 128. Check-mark indicates that the corresponding component is enabled. The best results are indicated as **bold** for each metric and dataset.

Dataset	Container	Bidirectional	Disc. ($\downarrow$)	Pred. ($\downarrow$)	cFID ($\downarrow$)	Corr. ($\downarrow$)
Energy	✓	✓	0.483 ± 0.050	75.519 ± 5.135	3.300	0.005 ± 0.161
			0.475 ± 0.053	71.788 ± 5.549	1.912	0.003 ± 0.227
	✓	✓	0.468 ± 0.033	70.819 ± 11.844	1.932	0.003 ± 0.036
			0.463 ± 0.018	68.870 ± 8.902	1.848	0.002 ± 0.026
ETTh	✓	✓	0.380 ± 0.106	0.609 ± 0.261	1.253	0.002 ± 0.141
			0.359 ± 0.090	0.569 ± 0.215	1.113	0.001 ± 0.185
	✓	✓	0.342 ± 0.093	0.565 ± 0.280	0.901	0.001 ± 0.209
			0.328 ± 0.085	0.524 ± 0.203	0.817	0.001 ± 0.081
LMT	✓	✓	0.475 ± 0.038	0.359 ± 0.134	0.186	0.068 ± 0.220
			0.333 ± 0.081	0.290 ± 0.099	0.146	0.062 ± 0.177
	✓	✓	0.341 ± 0.065	0.295 ± 0.068	0.155	0.068 ± 0.200
			0.267 ± 0.097	0.037 ± 0.013	0.125	0.050 ± 0.386

Table 3: Sensitivity analysis of the cycle-consistency loss weight λ_{cyc} on LMT and ETTh (sequence length 128). The best results are indicated as bold for each metric and dataset.

Dataset	λ_{cyc}	Disc. ($\downarrow$)	Pred. ($\downarrow$)	cFID ($\downarrow$)	Corr. ($\downarrow$)
LMT	0.0	0.378 ± 0.102	0.705 ± 0.471	0.890	0.061 ± 0.158
	0.3	0.350 ± 0.178	0.173 ± 0.050	0.225	0.043 ± 0.183
	0.6	0.267 ± 0.097	0.037 ± 0.013	0.125	0.050 ± 0.386
	1.0	0.467 ± 0.055	0.326 ± 0.149	0.251	0.058 ± 0.155
ETTh	0.0	0.413 ± 0.072	1.139 ± 0.261	1.368	0.012 ± 0.198
	0.3	0.394 ± 0.078	0.944 ± 0.889	0.846	0.005 ± 0.201
	0.6	0.328 ± 0.085	0.524 ± 0.203	0.817	0.001 ± 0.081
	1.0	0.388 ± 0.058	0.577 ± 0.576	0.920	0.017 ± 0.182

Table 4: Computational efficiency comparison averaged over the Energy, ETTh, and LMT datasets at sequence lengths $L=128$ and $L=256$. Parameter counts are reported in millions (M). Training time denotes the average wall-clock time required for model training, and inference time denotes the average generation time for 2,000 samples. All measurements are conducted under the same hardware and implementation setting.

Method	Params (M) $\downarrow$	Training Time $\downarrow$		Inference Time $\downarrow$	
		$L = 128$	$L = 256$	$L = 128$	$L = 256$
TimeVAE	0.33	1h 31m	3h 09m	2.9s	5.9s
TimeGAN	0.44	11h 55m	25h 40m	8.4s	17.6s
Diffusion-TS	0.60	6h 20m	17h 35m	6m 55s	20m 40s
TimeDiff	0.57	4h 42m	11h 18m	1m 12s	2m 43s
ImagenI2R	1.20	4h 37m	13h 58m	3m 14s	10m 48s
ImagenTime	1.13	3h 48m	11h 27m	2m 29s	8m 54s
TSFlow	0.18	1h 12m	2h 36m	3.4s	7.1s
FLOW-ITS	0.22	1h 02m	2h 14m	2.6s	5.4s

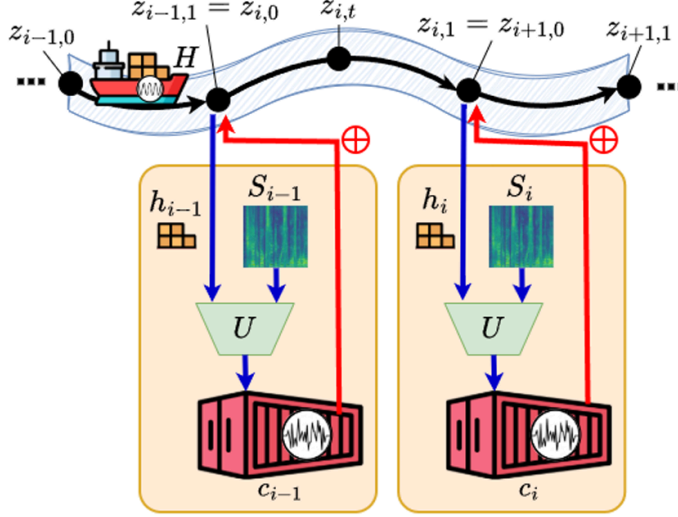


Figure 2: **Flow matching between adjacent samples.** The ship H guides the global transition of the latent sample $z_{i,t}$ between $z_{i,0}$ and $z_{i,1}$. Blue arrows indicate the unloading process by module U to generate container c_i . Red arrows represents the loading of the container to ship. The $\oplus$ symbol denotes concatenation.

Figure 4: Qualitative comparison between GT and generated data on LMT at a sequence length of 128. GT data are shown in red, and generated data are shown in blue.

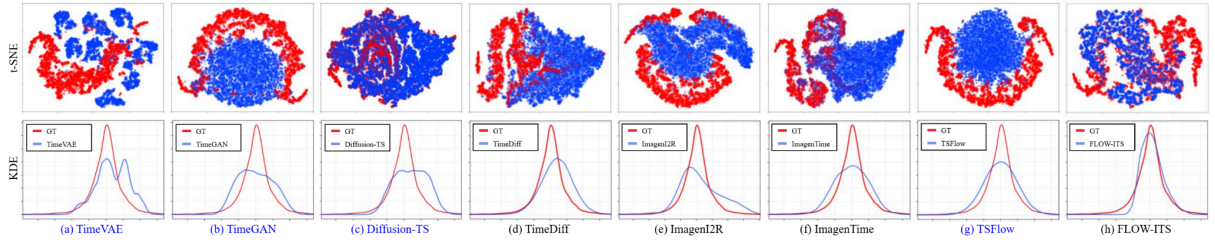
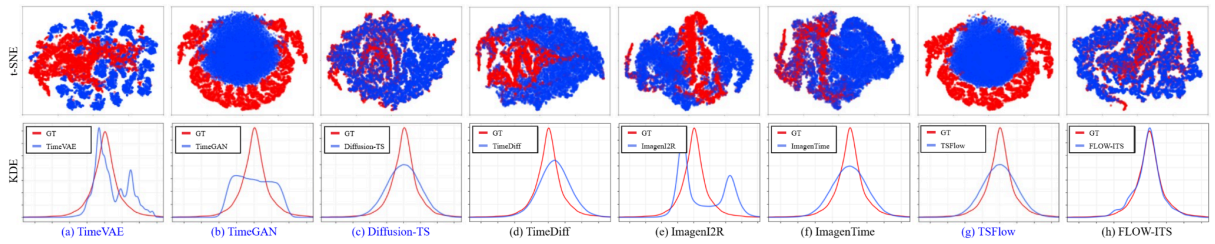


Figure 5: Qualitative comparison between GT and generated data on LMT at a sequence length of 256. GT data are shown in red, and generated data are shown in blue.



Algorithm 1. Test-time generation and reconstruction of FLOW-ITS

Input: Trained velocity network f_θ , container module U , decoder D , sequence length N

Output: Generated time series $\hat{x}$

1. Sample an initial latent ship $H^{(0)} = \{h_i^{(0)}\}_{i=0}^{N-1}$ from the latent prior.
2. Decode $H^{(0)}$ to obtain provisional spectrogram frames $\{\hat{S}_i^{(0)}\}_{i=0}^{N-1}$.
3. For each flow integration step:
 - a. Construct containers $c_i = U(\hat{S}_i, h_i)$ from generated frame estimates.
 - b. For each adjacent transition, compute the velocity $v_\theta(z_{i,t}, i, t) = f_\theta(z_{i,t}, i, t; H, c_i)$ conditioned on the full ship H .
 - c. Update the latent states and refresh the ship H .
4. Decode the final latent ship $\hat{H}$ into spectrogram frames $\{\hat{S}_i\}_{i=0}^{N-1} = D(\hat{H})$.
5. Convert the generated spectrogram sequence $\{\hat{S}_i\}_{i=0}^{N-1}$ to the time-series signal $\hat{x}$.
6. Return $\hat{x}$.